

# An Anatomy of Precision: The Epsilon–Delta Rigor of Limits and Continuity

A Mathematical Essay on Quantifiers, Inequalities, and Edge Cases

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## Abstract

The Cauchy–Weierstrass  $(\epsilon, \delta)$ -definitions of function limits and continuity are among the basic tools of analysis. While introductory calculus presents these definitions as standard technical formalisms, their small details matter: the quantifiers, inequalities, and exclusions are not interchangeable. This essay examines the definitions of function limits and continuity, with special attention to why  $0 < |x - a|$  is required for limits but absent in continuity, why strict inequalities are chosen over non-strict inequalities in quantifiers, and what goes wrong if these conditions are modified or relaxed.

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# 1 Introduction

In early calculus, a limit  $\lim_{x \rightarrow a} f(x) = L$  is often intuitively described as: “as  $x$  gets arbitrarily close to  $a$ ,  $f(x)$  gets arbitrarily close to  $L$ .” While visually appealing, this heuristic relies on kinematic concepts (“gets close to”, “approaches”) that invite logical paradoxes and ambiguity.

During the 19<sup>th</sup> century, Augustin-Louis Cauchy and Karl Weierstrass arithmeticized analysis by replacing dynamic motion with static logical quantifiers. The modern definitions convert topological proximity into precise real-number inequalities. However, it is easy to take the structural details of these definitions for granted. Why do we write  $0 < |x - a| < \delta$  instead of  $|x - a| < \delta$ ? Why do we require  $\epsilon > 0$  strictly? What breaks if we replace  $< \epsilon$  with  $\leq \epsilon$ ?

This essay analyzes these subtleties and shows that some small-looking changes alter the meaning of the definitions.

## 2 A Prototype: Equality by Arbitrary Precision

Before looking at limits, it is useful to isolate a simpler logical pattern that already contains the heart of  $(\epsilon, \delta)$  reasoning. The pattern is this: if a nonnegative quantity can be made smaller than every positive tolerance, then that quantity must be zero.

**Theorem 2.1** (Equality by Arbitrary Precision). *For any real numbers  $a, b \in \mathbb{R}$ ,*

$$a = b \iff \forall \epsilon > 0, |a - b| < \epsilon.$$

*Proof.* We prove both directions.

**Direction 1** ( $a = b \implies \forall \epsilon > 0, |a - b| < \epsilon$ ): Assume  $a = b$ . Then  $a - b = 0$ , so  $|a - b| = 0$ . Let  $\epsilon > 0$  be arbitrary. Since  $0 < \epsilon$ , we have  $|a - b| = 0 < \epsilon$ . Therefore

$$\forall \epsilon > 0, |a - b| < \epsilon.$$

**Direction 2** ( $\forall \epsilon > 0, |a - b| < \epsilon \implies a = b$ ): Assume

$$\forall \epsilon > 0, |a - b| < \epsilon.$$

We will show that  $a = b$ . Suppose, for contradiction, that  $a \neq b$ . Then  $|a - b| > 0$ . Choose

$$\epsilon_0 = \frac{|a - b|}{2}.$$

Since  $|a - b| > 0$ , we have  $\epsilon_0 > 0$ . By the assumed statement, applied to this particular positive tolerance  $\epsilon_0$ , we get

$$|a - b| < \epsilon_0 = \frac{|a - b|}{2}.$$

But this is impossible for a positive number  $|a - b|$ : it would imply

$$\frac{|a - b|}{2} < 0.$$

This contradiction shows that our supposition  $a \neq b$  was false. Hence  $a = b$ . □

This theorem explains why the phrase “for every  $\epsilon > 0$ ” is so powerful. It does not merely say that  $|a - b|$  is small. It says that no positive tolerance, however tiny, can still detect any separation between  $a$  and  $b$ . The only nonnegative distance with that property is zero.

The  $(\epsilon, \delta)$ -definition of a limit adds one extra layer to this prototype. Instead of directly proving that a fixed distance  $|a - b|$  is smaller than every  $\epsilon$ , it says:

$$\forall \epsilon > 0, \exists \delta > 0$$

such that whenever  $x$  is within the permitted  $\delta$ -neighborhood of  $a$ , the output distance  $|f(x) - L|$  is smaller than that chosen  $\epsilon$ . In other words, the  $\delta$  clause is a mechanism for forcing the output error below an arbitrary tolerance.

Thus a limit proof usually has the same skeleton as Theorem 2.1, but with a dependency inserted:

$$\text{given an arbitrary } \epsilon > 0, \quad \text{choose a suitable } \delta > 0, \quad \text{then prove } |f(x) - L| < \epsilon$$

for every  $x$  satisfying the domain and closeness conditions. The theorem above is the degenerate case where there is no input variable and no  $\delta$  to choose: the relevant distance is already fixed.

### 3 The Formal Definitions

Let  $D \subseteq \mathbb{R}$  be the domain of a real-valued function  $f : D \rightarrow \mathbb{R}$ , and let  $a \in \mathbb{R}$  be a limit point (or accumulation point) of  $D$ .

**Definition 3.1** (Limit of a Function). We say that the **limit of  $f(x)$  as  $x$  approaches  $a$  is  $L$** , denoted  $\lim_{x \rightarrow a} f(x) = L$ , if:

$$\forall \epsilon > 0, \exists \delta > 0, \forall x \in D, (0 < |x - a| < \delta \implies |f(x) - L| < \epsilon).$$

Now consider the definition of continuity at a point. Let  $a \in D$ .

**Definition 3.2** (Continuity at a Point). We say that  $f$  is **continuous at  $a$**  if:

$$\forall \epsilon > 0, \exists \delta > 0, \forall x \in D, (|x - a| < \delta \implies |f(x) - f(a)| < \epsilon).$$

Comparing Definition 3.1 and Definition 3.2, two structural differences immediately emerge:

1. **The explicit value of  $L$  vs.  $f(a)$ :** In the limit definition,  $L$  is a candidate value not necessarily tied to  $f(a)$  (indeed,  $f(a)$  need not even be defined). In continuity,  $f(a)$  is strictly required to exist, and  $L = f(a)$ .
2. **The lower bound  $0 < |x - a|$ :** The limit definition explicitly excludes  $x = a$ , whereas the definition of continuity includes  $x = a$ .

## 4 Excluding Zero: The Role of $0 < |x - a|$

The condition  $0 < |x - a|$  is equivalent to stating  $x \neq a$ . Its inclusion in the definition of a limit, and its omission in the definition of continuity, is not accidental.

### 4.1 Why Exclude $x = a$ in Limits?

The limit  $\lim_{x \rightarrow a} f(x)$  seeks to describe the *local behavior of  $f$  near  $a$* , completely independent of the *value of  $f$  at  $a$* .

Suppose we removed the constraint  $0 < |x - a|$  and defined a "modified limit"  $L'$  using:

$$|x - a| < \delta \implies |f(x) - L'| < \epsilon.$$

If  $x = a$  were allowed, then  $|a - a| = 0 < \delta$  holds for every  $\delta > 0$ . The implication would then require:

$$|f(a) - L'| < \epsilon \quad \forall \epsilon > 0.$$

Since the only non-negative real number less than every positive  $\epsilon$  is 0, this forces  $|f(a) - L'| = 0$ , meaning  $L'$  must equal  $f(a)$ .

This hypothetical alteration would immediately break standard examples in calculus:

- **Functions with removable discontinuities would have no limits.** Consider  $f(x) = \frac{x^2-1}{x-1}$  for  $x \neq 1$ , with  $f(1) = 5$ . Intuitively and standardly,  $\lim_{x \rightarrow 1} f(x) = 2$ . However, under the modified definition, no limit could exist because  $f(1) = 5 \neq 2$ .
- **Functions undefined at  $a$  could not have limits.** The standard derivative definition is:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}.$$

The difference quotient  $g(h) = \frac{f(a+h) - f(a)}{h}$  is explicitly undefined at  $h = 0$ . If  $h = 0$  were required in the limit quantifier, no derivative could ever be evaluated!

Thus, excluding  $x = a$  decouples local trend from point behavior.

## 4.2 Why Include $x = a$ in Continuity?

For continuity, the situation is reversed. Continuity asserts that the local trend *matches* the point value.

If we evaluate the continuity condition at  $x = a$ :

$$|a - a| < \delta \implies |f(a) - f(a)| < \epsilon \iff 0 < \delta \implies 0 < \epsilon.$$

For  $x = a$ , the statement  $|f(a) - f(a)| < \epsilon$  reduces to  $0 < \epsilon$ , which is vacuously true for all  $\epsilon > 0$ . Thus, explicitly writing  $0 < |x - a| < \delta$  or  $|x - a| < \delta$  in the definition of continuity produces logically equivalent statements. Including  $x = a$  simply simplifies the notation and emphasizes that  $f(a)$  must be defined.

## 5 Strict vs. Non-Strict Inequalities in Quantifiers

A subtle point of confusion for students is why we use strict inequalities ( $\epsilon > 0$ ,  $\delta > 0$ ,  $|x - a| < \delta$ ,  $|f(x) - L| < \epsilon$ ) rather than weak inequalities ( $\geq, \leq$ ). Does changing strict inequalities to weak ones change the mathematical meaning?

Let us test all permutations systematically.

### 5.1 Can we replace $|f(x) - L| < \epsilon$ with $|f(x) - L| \leq \epsilon$ ?

**Theorem 5.1.** *The definition of limit with  $|f(x) - L| \leq \epsilon$  is equivalent to Definition 3.1.*

*Proof.* Let Definition A be the standard definition using  $< \epsilon$ , and Definition B be the modified definition using  $\leq \epsilon$ .

**Direction 1 (A  $\implies$  B):** Assume  $\lim_{x \rightarrow a} f(x) = L$  under Definition A. Given  $\epsilon > 0$ , by Definition A there exists  $\delta > 0$  such that  $0 < |x - a| < \delta \implies |f(x) - L| < \epsilon$ . Since  $|f(x) - L| < \epsilon$  implies  $|f(x) - L| \leq \epsilon$ , Definition B holds with the same  $\delta$ .

**Direction 2 (B  $\implies$  A):** Assume Definition B holds. Given  $\epsilon > 0$ , choose  $\epsilon' = \frac{\epsilon}{2} > 0$ . By Definition B, there exists  $\delta > 0$  such that  $0 < |x - a| < \delta \implies |f(x) - L| \leq \epsilon' = \frac{\epsilon}{2}$ . Since  $\frac{\epsilon}{2} < \epsilon$ , we have  $|f(x) - L| < \epsilon$ . Thus Definition A holds.  $\square$

Hence, using  $\leq \epsilon$  instead of  $< \epsilon$  changes nothing mathematically. The strict inequality  $< \epsilon$  is preferred purely for aesthetic symmetry with  $|x - a| < \delta$ .

## 5.2 Can we replace $0 < |x - a| < \delta$ with $0 < |x - a| \leq \delta$ ?

**Theorem 5.2.** *The definition of limit with  $0 < |x - a| \leq \delta$  is equivalent to Definition 3.1.*

*Proof.* If a  $\delta_1 > 0$  satisfies the strict inequality condition, taking  $\delta_2 = \frac{\delta_1}{2} > 0$  satisfies the weak inequality condition because  $0 < |x - a| \leq \delta_2 \implies 0 < |x - a| < \delta_1$ . Conversely, if  $\delta_1$  works for weak inequality, it automatically works for strict inequality since  $0 < |x - a| < \delta_1 \implies 0 < |x - a| \leq \delta_1$ .  $\square$

## 5.3 What if we allow $\epsilon = 0$ ( $\epsilon \geq 0$ )?

What happens if we modify the quantifier from  $\forall \epsilon > 0$  to  $\forall \epsilon \geq 0$ ?

Suppose we require:

$$\forall \epsilon \geq 0, \exists \delta > 0, 0 < |x - a| < \delta \implies |f(x) - L| \leq \epsilon.$$

Setting  $\epsilon = 0$ , the condition demands that there exists  $\delta > 0$  such that for all  $x$  in the punctured neighborhood  $0 < |x - a| < \delta$ :

$$|f(x) - L| \leq 0 \implies f(x) = L.$$

This forces  $f(x)$  to be *strictly constant* in an open punctured neighborhood around  $a$ .

Thus, allowing  $\epsilon = 0$  makes the definition far too strong. Non-constant functions like  $f(x) = x$  would fail to have a limit anywhere! The strict positivity  $\epsilon > 0$  is what allows  $f(x)$  to approximate  $L$  without forcing immediate equality.

## 5.4 What if we allow $\delta = 0$ ( $\delta \geq 0$ )?

Suppose the definition allowed choosing  $\delta = 0$ . If  $\delta = 0$ , the condition  $0 < |x - a| < 0$  is a contradiction—no real number  $x$  can satisfy it. The antecedent of the implication:

$$0 < |x - a| < 0 \implies |f(x) - L| < \epsilon$$

becomes *vacuously true* for *every* real number  $L$  and every function  $f$ .

If  $\delta = 0$  were allowed, every function would have *every* real number as its limit at every point, so limits would no longer be unique. Hence,  $\delta$  must be strictly positive ( $\delta > 0$ ).

# 6 Summary Matrix of Variant Definitions

To synthesize these logical dependencies, the table below summarizes the effect of altering individual components of the  $(\epsilon, \delta)$ -definition for  $\lim_{x \rightarrow a} f(x) = L$ :

| Modification                                        | Mathematical Consequence                        | Status             |
|-----------------------------------------------------|-------------------------------------------------|--------------------|
| Replace $< \epsilon$ with $\leq \epsilon$           | Equivalent.                                     | Valid alternative. |
| Replace $< \delta$ with $\leq \delta$               | Equivalent.                                     | Valid alternative. |
| Remove $0 <  x - a $                                | Forces $L = f(a)$ . Breaks derivatives.         | Invalid.           |
| Allow $\epsilon \geq 0$                             | Forces $f(x) = L$ locally (constant function).  | Invalid.           |
| Allow $\delta \geq 0$                               | Vacuously true for all $L$ . Limits not unique. | Invalid.           |
| Swap quantifiers: $\exists \delta \forall \epsilon$ | Forces $f(x) = L$ identically near $a$ .        | Invalid.           |

Table 1: Impact of small formal changes on the limit definition.

## 7 Conclusion

The  $(\epsilon, \delta)$ -definition of a limit is carefully tuned. The inclusion of  $0 < |x - a|$  separates the limit from the function's value at the point, which is what makes removable discontinuities and derivative quotients possible. The strict positivity of  $\epsilon$  and  $\delta$  leaves room for approximation without collapsing the definition into exact equality or vacuous truth. Choices that can look like minor notational preferences are often doing real logical work.